

DIGITAL – TWIN BASED SMART REEFER CONTAINER WITH EARLY FAULT DETECTION

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I. ABSTRACT:

Refrigerated containers (reefers) are critical assets in cold-chain logistics, where maintaining strict environmental conditions is essential to prevent cargo spoilage. Conventional reefer monitoring systems are largely reactive, relying on threshold-based alarms that activate only after operating limits are violated.

This work proposes a “Digital – Twin” based Smart Reefer Container integrated with Random Forest Classifier machine learning model to analyse temporal trends for Early Fault detection and predictive monitoring.

The proposed model demonstrates a Data Acquisition Dashboard that monitors the Smart Reefer while simulating faults in it to serve as a platform for both operator training and machine-learning dataset generation.

Keywords: Smart Reefer; Digital Twin; Predictive Maintenance; Random Forest Classifier; Data Acquisition Dashboard; Cold-Chain Logistics

II. INTRODUCTION:

Reefer containers are sensitive to any deviation from prescribed environmental conditions which can result in irreversible cargo damage and significant economic loss. Existing reefer monitoring solutions typically focus on real-time measurement and alarm generation once predefined thresholds are exceeded.

With advancements in digital twins and machine learning, it is now possible to move beyond reactive systems toward intelligent, predictive monitoring frameworks. This project explores the application of these technologies in the context of smart reefer container monitoring.

III. OBJECTIVE:

The primary objectives of the proposed model are:

1. To develop a digital twin simulation of a reefer container that mirrors real-time physical behaviour.
2. To enable interactive fault injection for simulating real-world failure scenarios.
3. To assess cargo spoilage risk based on cumulative environmental exposure.
4. To implement machine learning based early fault detection for predictive monitoring.
5. To generate ML-ready datasets for future research and model improvement.

IV. NOVELTY/ CONTRIBUTION:

The novelty of the proposed model lies in the integration of multiple advanced concepts into a single real-time platform:

- Use of a digital twin rather than a static simulation.
- Early fault detection based on trend analysis instead of threshold violations.
- A hybrid physics and machine learning architecture with physical guardrails.
- Clear separation between fault prediction (cause) and cargo risk assessment (effect).
- A unified system capable of data generation, prediction, and visualization.

This integrated approach is rarely implemented at the academic model-making level.

V. WORKING PRINCIPLE:

The digital twin based smart reefer that combines physics-based simulation, interactive fault injection, cargo risk assessment, and machine learning based early fault prediction in a single real-time platform. The architecture of the simulation is illustrated below:

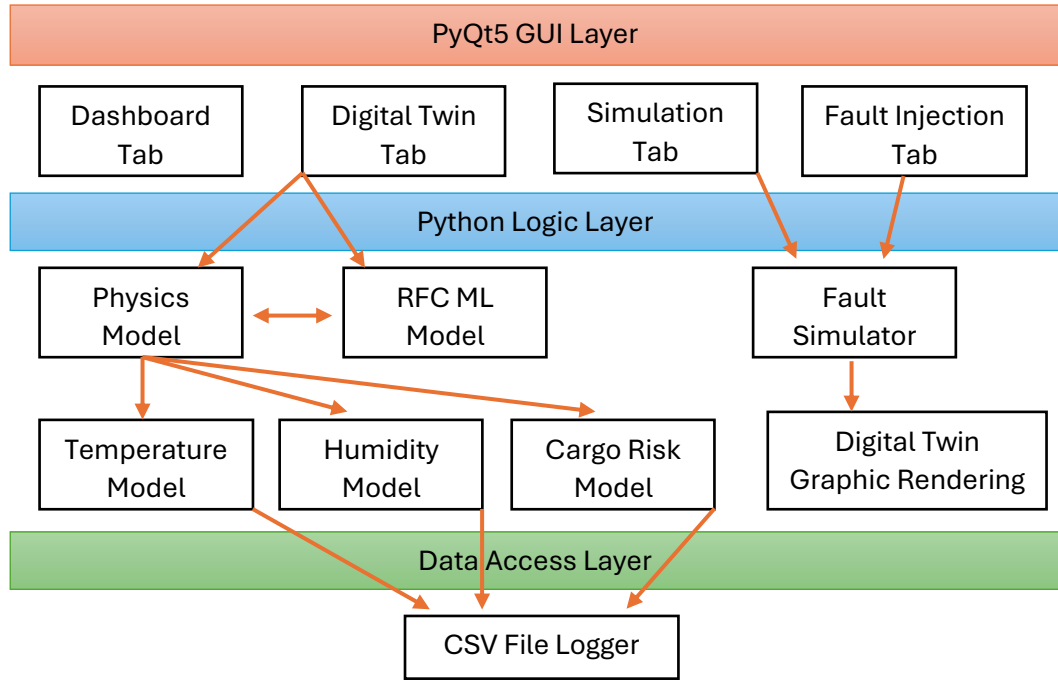


Fig.1: System Architecture of the Digital Twin Simulation Model [1]

The mathematical models used to simulate the physics of temperature, humidity and cargo spoilage risk are: (Discrete Time Implementation)

1. 1st Order Lumped Thermal Model (Newtonian Heating/Cooling):

$$T_{k+1} = T_k + k(T_{amb} - T_k)\Delta t - u_k C \Delta t \quad (1)$$

2. 1st Order Dynamic Moisture Ingress Model:

$$H_{k+1} = H_k + \alpha(H_{target} - H_k)\Delta t \quad (2)$$

$H_{target} = H_{door}$ (door open), $H_{nominal}$ (door closed)

3. Cumulative Thermal Exposure Risk Model:

$$R_{k+1} = R_k + \beta \max(0, T_k - T_{safe}) \Delta t \quad (3)$$

$$T_{safe,low} \leq T(t) \leq T_{safe,high}$$

The Random Forest Classifier ML Model predicts the fault based on the vectors of temperature, moisture, risk and standard deviation of temperature and humidity compared to an early fault detection threshold as follows:

4. Early Fault Prediction Model:

$$\text{Early Fault} = 1 \text{ if } P_{fault} \geq \theta, \text{ else } 0$$

$$x(\text{vectors}) = [\bar{T}, \Delta T/\Delta t, \bar{H}, \Delta H/\Delta t, \Delta R/\Delta t, \sigma_T, \sigma_H] \quad (4)$$

The simulation works in the following manner:

- Operational states such as power availability, door status, and cooling system health are maintained in real time.
- Faults can be induced manually to simulate real-world conditions.
- Temperature and humidity trends are logged and processed through a machine learning model trained to predict faults several minutes in advance.
- A cargo spoilage risk index accumulates based on prolonged exposure to unfavourable conditions.
- Visual indicators on the dashboard distinguish between predictive fault warnings and actual cargo risk.

VI. DETAILED DIAGRAM:

The following image illustrates the simulation of the digital twin in an ideal condition:

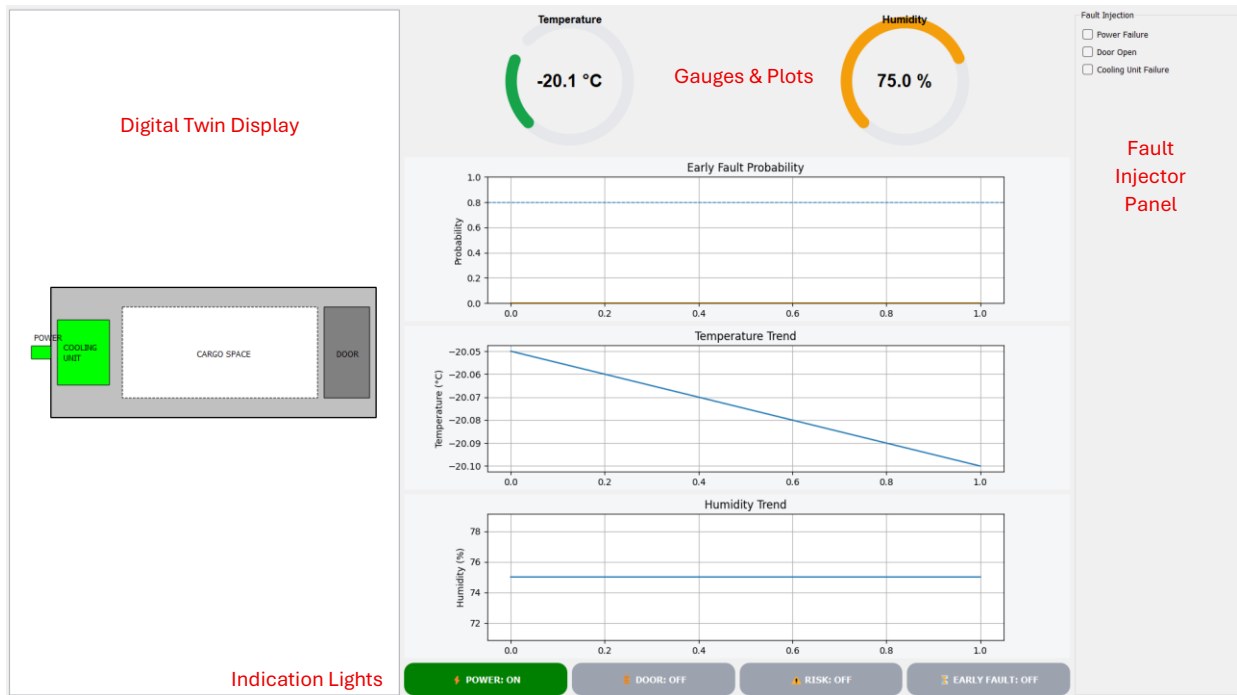


Fig.2: Dashboard of Digital Twin Simulation of a Reefer Container [2]

VII. RESULTS:

The following cases have been simulated and verified via graphs and ML Model:

- **Case I: Single Unit Failure (Cooling Unit Failure)**

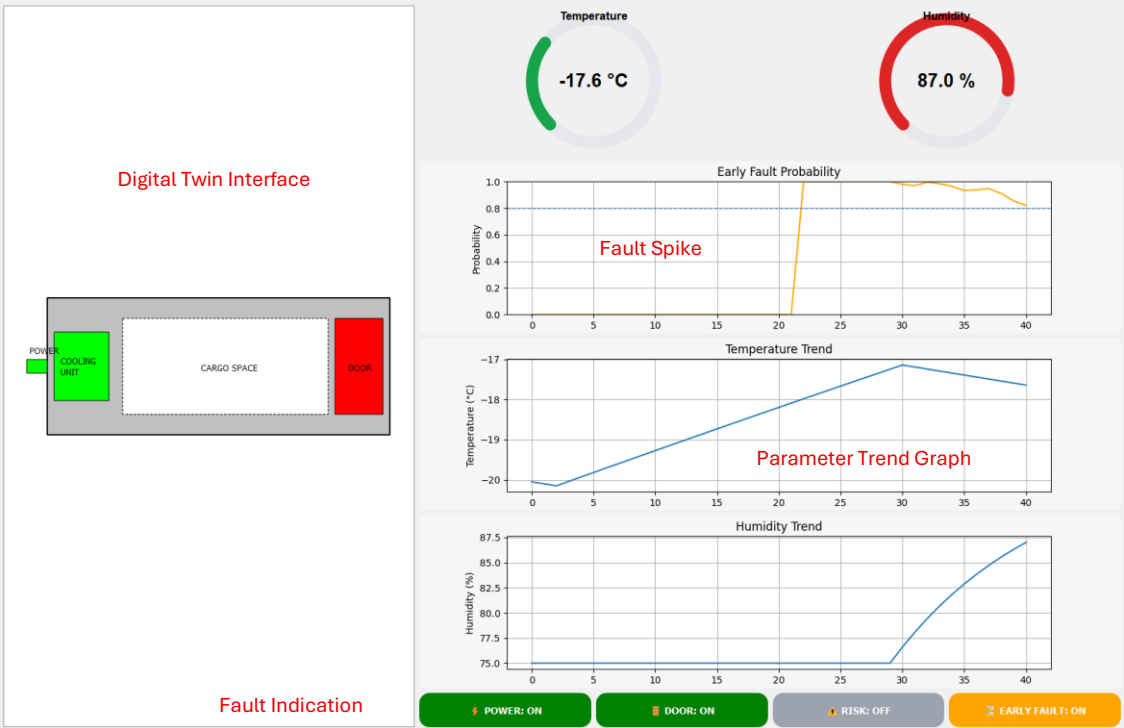


Fig.3: Single Unit Failure Simulation [2]

- **Case II: Multi failure (Door Failure + Cooling Unit Failure)**

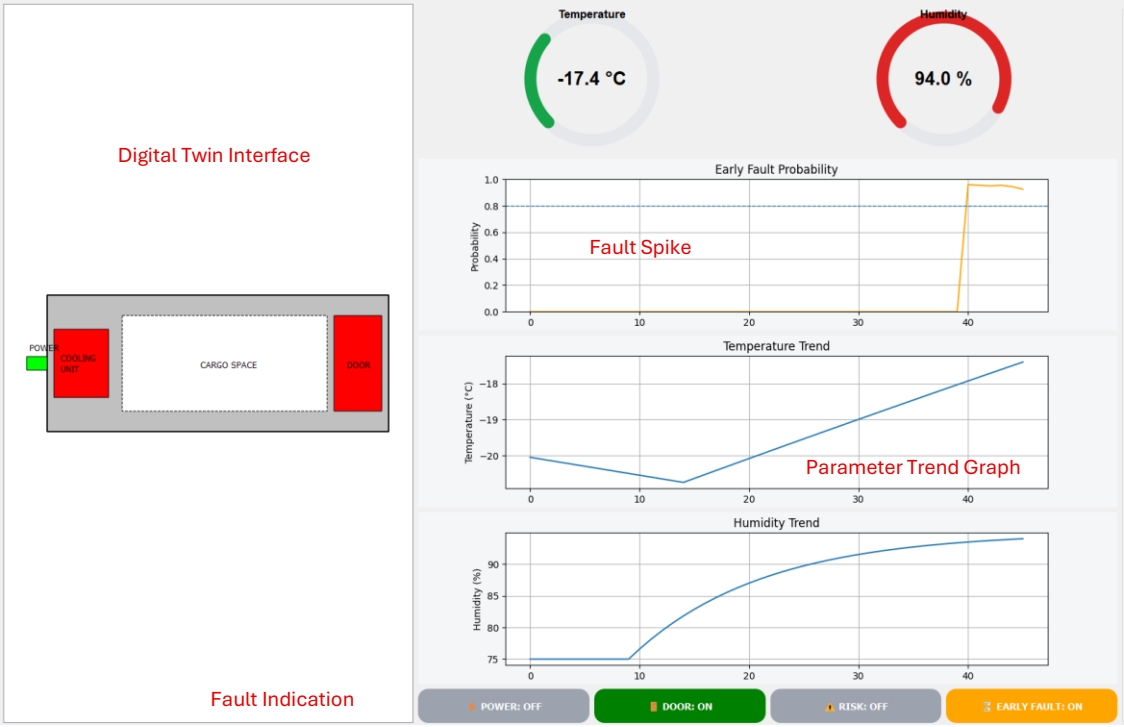


Fig.4: Multi Unit Failure Simulation [2]

- **Case III: Cargo Spoilage (Risk Light Indication)**

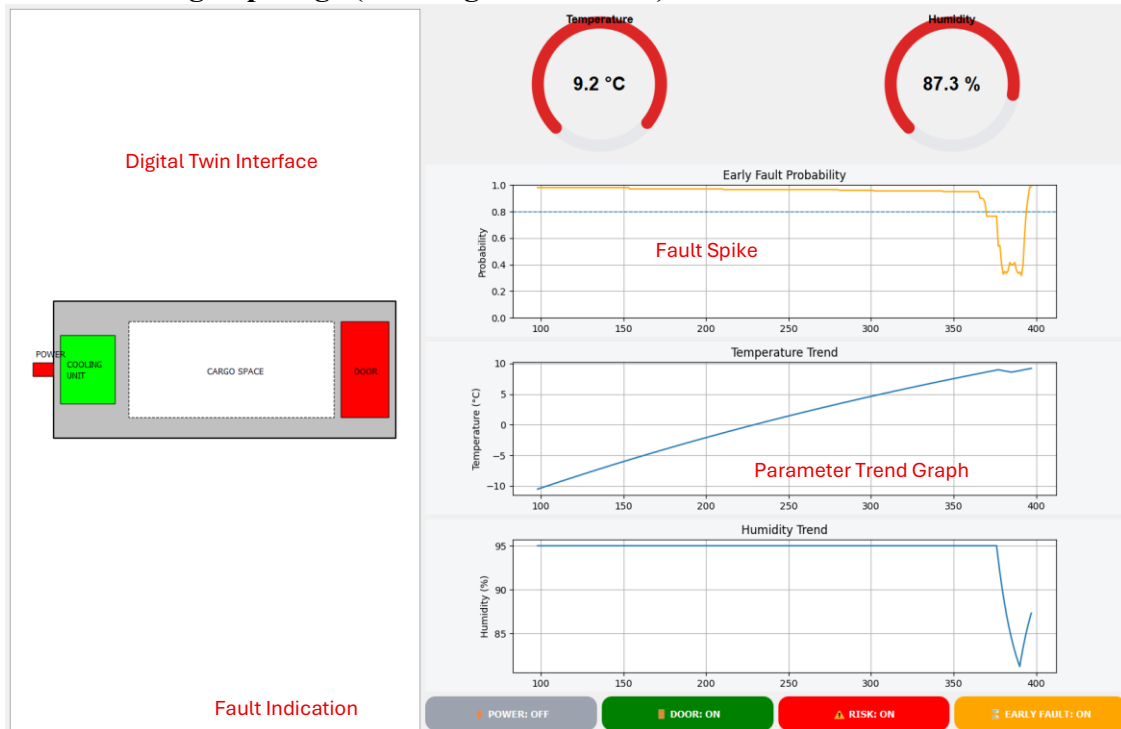


Fig.5: Cargo Spoilage Indication along with Fault simulation [2]

VIII. CONCLUSION:

The proposed digital twin based smart reefer container successfully demonstrates predictive monitoring through machine learning based early fault detection. By combining physics-based modelling, interactive simulation, and data-driven intelligence, the system shifts reefer monitoring from a reactive to a preventive paradigm.

This work highlights the potential of digital twins and machine learning in improving reliability and safety in cold-chain logistics.

IX. REFERENCES:

- [1] T. F., "Digital Twin in Industry: State-of-the-Art," *IEEE Transactions on Industrial Informatics*, 2009.
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